Arjun Nanduri

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Education

University of California: Berkeley, BA in Computer Science — Expected May 2028

- GPA: 3.9 | SAT: 1590
- Courses: Probability & Stochastic Processes, Machine Learning, Macroeconomics, Data Structures

Experience

Machine Learning Engineer, Aver Technologies Inc. – Virginia — Aug 2025 – Present

- Conducted field interviews with construction workers to surface real-world constraints — inconsistent hand-drawn markings, Spanish-speaking workforce — **directly changed model design decisions**.
- Building a multi-stage PyTorch CV pipeline to automate depth tracking when driving piles (foundational support for heavy structures, like bridges) into the ground — **replacing a manual, hand-marked DOT compliance process**.
- Designing two-phase roadmap: Phase 1 CV digitization of manual process, **Phase 2 LIDAR elimination of markings**.

Software Engineering Intern, CACI International Inc. – Virginia — May 2025 – Aug 2025

- Built a full-stack logistics web app for **U.S. Customs field officers** using Angular, Spring Boot, and PostgreSQL.
- Expanded database test coverage **by 20%**, enhancing system stability and enabling more efficient bug detection.
- **Ramped quickly onto unfamiliar enterprise Spring Boot codebase** in Agile team, contributing through sprint planning, standups, and code reviews.

Data Science Research Intern, CDSS, UC Berkeley – Berkeley, CA — Jan 2025 – May 2025

- Stepped up to lead disorganized team with no formal authority — **built shared task system**, assigned work by strength, held mock presentations, presented at UC Berkeley Discovery Symposium.
- Fine-tuned a BART transformer for low-resource NLP, **beating open-source baselines by 15%**.
- Owned the data pipeline for a ∼1M-word, ∼10K-text raw corpus — preprocessing and alignment work that **had to be right before any model result was credible**.

Projects

PotBot — Civic Reporting Agent, *CalHacks Prize Winner* — https://devpost.com/software/potbot

- Led a team of 3 under a 36-hour hackathon deadline, **prioritizing a working demo over an unfinished feature set** — shipped an agentic pipeline (Llama vision, Playwright) that auto-submits real infrastructure repair requests to SF city government, winning CalHacks' sponsor prize.

Quadruped Locomotion with PPO (MuJoCo, Gymnasium, SB3) — github.com/FavouritePlayer/ant_sim

- Validated a leg-damage-specialist policy across multiple seeds and found it didn't transfer to other legs (100% fall rate) — **killed that direction rather than ship an unsupported claim**, building a compound policy router instead.
- Terrain-adapted policy, validated across 10 matched seeds, **achieved 2.1x the flat-baseline reward** on unseen terrain — the approach that shipped.

DealScout — Resale Arbitrage Agent (LangGraph, Playwright) — github.com/FavouritePlayer/DealScout

- Built a LangGraph-orchestrated agent that scrapes live Craigslist listings via Playwright, scores resale value with deterministic arithmetic kept separate from LLM judgment, then restricted image-attachment to post-filter candidates to **avoid a ∼4x blowup in scan time**.

Context Custodian — Workspace Hygiene Agent (FastAPI) — github.com/FavouritePlayer/ContextCustodian

- Built a FastAPI agent that compacts and cleans an agent-first workspace for downstream agents to ingest — a three-stage prompt-injection detection pipeline (regex, vector similarity, LLM adjudication) with a fail-open fallback, where **every fix is a reversible move, never a delete**.

Face Morpher — From-Scratch Warping (NumPy, dlib, scipy) — github.com/FavouritePlayer/face-morpher

- Chose to implement image warping from scratch in NumPy — affine solve, inverse mapping, bilinear interpolation — instead of relying on OpenCV's built-in warp, **restricting OpenCV to I/O only** to own geometric precision outright.

Skills

Languages: Python, Java, JavaScript, SQL

Technologies: React.js, Angular, Spring Boot, Flutter, AWS, PyTorch, Git